Pillzilla — Project Documentation

Pillzilla is a medication reminder Android app (Java, minSdk 26, Material 3). Users register/log in with Firebase Auth, add medications with a schedule and repeat frequency, get reminded via `AlarmManager` + system notifications, and can mark doses taken. This document covers how it's built, why it's built that way, and what to watch out for.

1. Architecture overview

```
MainActivity (router, no UI)
 |- not logged in -> LoginScreen <-> RegisterScreen
 `- logged in     -> HomeActivity
                      |- Toolbar (shared, title changes per tab)
                      |- fragmentContainer (swaps fragments, no Navigation Component)
                      |   |- HomeFragment          (medication list, search, filter, add)
                      |   |- NotificationsFragment  (notification history, clear all)
                      |   `- ProfileFragment        (avatar, edit name, logout, delete account)
                      `- BottomNavigationView (drives the fragment swap)

AddingMedication (separate Activity, opened from HomeFragment's FAB)
MedicationReminderReceiver (BroadcastReceiver, fired by AlarmManager - this is where reminders actually
happen)
```

There's no ViewModel/Repository layer, no dependency injection, and no RecyclerView — this is a deliberately simple, beginner-scaled codebase (see §4 Decisions). Each screen talks to Firebase directly.

Navigation is manual `FragmentTransaction.replace()` calls in `HomeActivity`'s `BottomNavigationView.OnItemSelectedListener` — there's no Jetpack Navigation graph. Every fragment switch creates a **new** fragment instance (not cached/retained), so each tab reloads its Firebase listener on every visit.

2. Data model (Firebase Realtime Database)

No local database (Room, SQLite) — Firebase Realtime Database is the single source of truth, read via live `ValueEventListener`s (so the UI updates automatically when data changes anywhere).

```
users/{uid}              -> User                 { name, email }
medications/{uid}/{id}   -> Medication            { id, name, dosage, scheduleTime, date, repeat, notes,
taken }
notifications/{uid}/{id} -> MedicationNotification { id, medicationId, title, message, time }
```

- `Medication.repeat` is one of `"Daily" | "Weekly" | "Monthly" | "Once"` (free-text string, not an enum).

- `MedicationNotification.medicationId` links a notification back to the medication that triggered it — this was **missing originally** and had to be added (see §5 Challenges) so that tapping a notification can open that medication's detail dialog.
- All three model classes need a public no-arg constructor for Firebase's `DataSnapshot.getValue(Class)` deserialization.

3. File-by-file reference

Activities

File	Role
`MainActivity`	Pure router. Checks `FirebaseAuth.getCurrentUser()` and immediately forwards to `HomeActivity` or `LoginScreen`, then `finish()` s itself. Also calls `NotificationHelper.createNotificationChannel()` once at startup.
`LoginScreen`	Email/password sign-in via `FirebaseAuth.signInWithEmailAndPassword`. Links to `RegisterScreen`.
`RegisterScreen`	Creates the Firebase Auth user, then writes a `User` record to `users/{uid}`. Links back to `LoginScreen`.
`HomeActivity`	Hosts the toolbar + bottom nav + fragment container. Sets the toolbar title per tab ("My Medications" / "Notifications" / "Profile"). Requests `POST_NOTIFICATIONS` permission on API 33+.
`AddingMedication`	Form to create a medication: name, dosage, date (`MaterialDatePicker`), time (`TimePickerDialog`), repeat (`AutoCompleteTextView` dropdown), notes. On save: writes to Firebase **and** schedules the first `AlarmManager` reminder.

Fragments (all hosted inside `HomeActivity`)

File	Role
`HomeFragment`	Medication list (`ListView` + `MedicationAdapter`), search box, taken/not-taken/all filter (toolbar dropdown), FAB -> `AddingMedication`, tap-to-edit dialog.
`NotificationsFragment`	Notification history list, "Clear All" (with confirmation), tap a notification -> looks up its linked medication and opens the same detail dialog as Home.

File	Role
`ProfileFragment`	Avatar + name + email header, edit username dialog, logout, delete account (wipes `users/`, `medications/`, `notifications/` for that uid, then deletes the Firebase Auth user).

Adapters / models

File	Role
`MedicationAdapter`	`ArrayAdapter<Medication>` for the Home list. No ViewHolder pattern — `findViewById` runs on every `getView()` (fine at this list size; see §4).
`NotificationAdapter`	`ArrayAdapter<MedicationNotification>`, takes an `OnNotificationClickListener` so `NotificationsFragment` can react to taps.
`Medication`, `MedicationNotification`, `User`	Plain POJOs for Firebase (de)serialization.

Reminders / notifications

File	Role
`MedicationReminderReceiver`	`BroadcastReceiver` fired by `AlarmManager`. On each fire: posts the system notification, saves a `MedicationNotification` to Firebase (with `medicationId`), resets that medication's `taken` flag to `false`, and — if `repeat != "Once"` — schedules the **next** alarm. This is the entire repeat engine; see §6.
`NotificationHelper`	Creates the Android notification channel (`medication_reminders`), called once from `MainActivity`.

Theming / shared resources

File	Role
`values/themes.xml`, `values-night/themes.xml`	`Base.Theme.Pillzilla` (parent `Theme.Material3.DayNight.NoActionBar`). Defines `colorPrimary` (different purple shade per mode), the app-wide font (`android:fontFamily` / `fontFamily` -> `@font/montserrat`), the shared rounded `TextInputLayout` style (`Widget.Pillzilla.TextInputLayout`, 16dp corners — matches card corners), and the destructive-button style (`Widget.Pillzilla.Button.Destructive`, uses `?attr/colorError` / `?attr/colorOnError`).

File	Role
`font/montserrat.xml`	Font-family with exactly two weights (400 regular, 700 bold) — see §5.
`layout_toolbar.xml`	Reusable `MaterialToolbar`, `<include>`d by both `activity_home.xml` and `activity_adding_medication.xml` so every screen's toolbar looks identical.
`item_medication.xml`, `item_notification.xml`	The two list-row cards — deliberately built to the same visual spec (`MaterialCardView`, 16dp corners, 1dp stroke, tonal icon circle) so the two lists feel like one system.
`menu_home_filter.xml`, `spinner_icon_toolbar.xml`	The Home-only toolbar filter control — see §6.

4. Key decisions

- **`ListView` / `ArrayAdapter`, not `RecyclerView`.** No `androidx.recyclerview` dependency exists in `build.gradle.kts`. This project intentionally stays on the older, simpler list API rather than introducing ViewHolder patterns, `DiffUtil`, etc. — appropriate for the project's scale and skill level.
- **No repository/shared-utility layer.** Every fragment/activity constructs its own `FirebaseDatabase.getReference(...)` calls and duplicates near-identical boilerplate (e.g. `HomeFragment` and `NotificationsFragment` each have their own copy of `showMedicationDialog()` / `cancelReminder()`). This is consistent with the rest of the codebase's style — three duplicated methods across two files was judged simpler than introducing a shared class for a two-fragment app.
- **Firebase Realtime Database + live listeners, no local cache.** Every list screen calls `addValueEventListener` and re-renders on any remote change. Simple, but means the app does nothing useful offline.
- **`MenuProvider` for the Home filter**, not a fragment-hosted `Spinner` in the layout. Scoped to `getViewLifecycleOwner()` + `Lifecycle.State.RESUMED` so the filter only appears in the toolbar while Home is the active tab — the modern, lifecycle-safe replacement for the deprecated `setHasOptionsMenu` / `onCreateOptionsMenu` pair.
- **One shared `Widget.Pillzilla.TextInputLayout` style** instead of repeating `boxCornerRadius*` on every field, and **one shared `Widget.Pillzilla.Button.Destructive` style** instead of repeating `backgroundTint` / `textColor` on every delete button — set once in `themes.xml`, referenced by name, so the corner radius / destructive color only needs to change in one place.
- **`?attr/colorError` / `?attr/colorOnError` for destructive actions**, not a hardcoded red hex — this is a real Material 3 color role, already theme-aware (adapts light/dark) without any extra work.
- **Icons are outlined, not filled**, across the whole app (notification bell, profile person icon, filter icon) — established after the filled variants were flagged as visually "out of place" against the rest of the outline-style icon set.

5. Challenges encountered & fixed

These were real bugs found and corrected during development — noted here so the reasoning isn't lost:

1. **Date picker was off by one day.** `MaterialDatePicker` always returns its selection as milliseconds at **UTC midnight**. The original code formatted that with a `SimpleDateFormat` that had no explicit timezone, so it fell back to the device's local timezone — for any timezone behind UTC, UTC-midnight-on-the-23rd renders as the evening of the 22nd. **Fix:** `sdf.setTimeZone(TimeZone.getTimeZone("UTC"))` before formatting the selected date, in `AddingMedication.showDatePicker()`.
2. **Notifications had no link back to their medication.** `MedicationNotification` originally only stored `title` / `message` / `time` — there was no way to know which medication a notification was about, which blocked "tap a notification to open its medication." **Fix:** added `medicationId` to `MedicationNotification`, threaded through from `MedicationReminderReceiver.saveNotificationToFirebase()`. Notifications saved *before* this change won't have it, so `NotificationsFragment` shows a "not linked to a medication" toast instead of crashing on old data.
3. **Toolbar existed but was effectively invisible.** `HomeActivity` already called `setSupportActionBar()`, but nothing ever called `setTitle()`, so every tab showed the same static app-label text. **Fix:** `HomeActivity` now sets the toolbar title at the same point it swaps fragments.
4. **Inconsistent spacing/corner radii read as "unpolished."** The search box used a `12dp` margin while list cards used `16dp` (edges didn't line up), and `TextInputLayout` fields used the Material 2 default ~4dp corner radius while cards used 16dp. **Fix:** aligned the search box margin to the cards' 16dp, and added the shared `Widget.Pillzilla.TextInputLayout` style so every text field matches the card/button corner radius.
5. **Added font files broke the build outright.** The `.ttf` files dropped into `res/font/` had uppercase letters and hyphens (`Montserrat-Regular.ttf`), which Android's resource compiler rejects (file-based resource names must be `[a-z0-9_.]+`). **Fix:** kept only Regular + Bold, renamed to `montserrat_regular.ttf` / `montserrat_bold.ttf`, and removed the other 16 unused weight/italic variants.
6. **Toolbar `Spinner` needed to show an icon, not the selected option's text.** A stock `ArrayAdapter` always renders the selected item's text in the collapsed view. **Fix:** the adapter used for the filter spinner overrides `getView()` to unconditionally inflate a fixed icon layout (`spinner_icon_toolbar.xml`), while `getDropDownView()` (used for the actual popup list) is left showing normal text rows.

6. Core logic: repeat & taken status

This is worth spelling out explicitly since it's easy to assume "marking taken" affects scheduling — **it doesn't**.

- **Scheduling is alarm-driven, not app-driven.** `AddingMedication.scheduleReminder()` sets exactly one `AlarmManager.setExactAndAllowWhileIdle()` alarm for the chosen date/time. When that alarm fires, `MedicationReminderReceiver.onReceive()`:
    1. Posts the system notification and saves a `MedicationNotification` row (with `medicationId`).
    2. Resets that medication's `taken` field to `false` in Firebase.
    3. If `repeat != "Once"`, computes `Calendar.getInstance()` (i.e. **now**, the actual fire time) `+ 1 day/week/month` and re-arms the next alarm via `rescheduleIfRepeating()`.

- **The "taken" switch never touches `AlarmManager`.** Both `HomeFragment` and `NotificationsFragment`'s detail dialogs write `taken` straight to Firebase (`medicationsRef.child(id).child("taken").setValue(isChecked)`) and nothing else. The next reminder fires at whatever time was already scheduled regardless of whether you mark the current dose taken.
- **Known drift risk:** because step 3 above reschedules from "now" rather than the original target hour/minute, a late-firing alarm (Doze deferral, device off, etc.) will push every subsequent occurrence's time-of-day later — repeats aren't pinned back to the original scheduled time. Not yet fixed; flagged here for awareness.
- **The Home screen's filter dropdown** (`HomeFragment`, toolbar) is a pure display filter over `taken` — "All / Not Taken / Taken" — combined with the search box text. It doesn't touch Firebase or scheduling at all, just what's shown in `displayedList`.

7. Known loose ends / dead resources

Found while documenting — not urgent, but worth knowing about:

- **`app/src/main/res/fonts/`** (plural) contains a full, unused Google Fonts download (variable fonts, `static/` folder with all 18 Montserrat weights, `OFL.txt`, `README.txt`). It's a different directory name than the one Android actually reads (`res/font/`, singular), so none of it is compiled into the app — it's just sitting in the repo. Left in place since it wasn't in scope to delete; safe to remove if you don't need it as a source archive.
- **`create_account.xml`** and **`activity_main.xml`**'s `textView5` are unused/placeholder leftovers (`create_account.xml` isn't referenced by any Activity/Fragment; `activity_main.xml` is used only as `MainActivity`'s content view, but that screen never actually renders anything visible since `MainActivity` redirects and finishes immediately).
- **`values/Repeat.xml`** defines an `<array name="Repeat">` (`Daily`, `Weekly` only) that's never referenced — `AddingMedication` uses its own hardcoded `String[] repeatOptions = {"Daily", "Weekly", "Monthly", "Once"}` instead.
- **`values/spacing.xml`**'s `marginTopList` dimen is unreferenced.
- **Unused drawables:** `chip_background.xml`, `outline_dehaze_24.xml`, `outline_density_medium_24.xml`, `baseline_ac_unit_24.xml` — none are referenced anywhere in layouts or code.
- **`menu_home.xml`** is a dead leftover menu resource from before the bottom-nav/fragment refactor (superseded by `menu_bottom_nav.xml`); nothing inflates it.

8. Possible future improvements

Not implemented, but natural next steps if the project grows:

- Fix the repeat-scheduling drift noted in §6 (pin reschedule to the original hour/minute, not "now").
- Migrate `ListView` / `ArrayAdapter` -> `RecyclerView` if the medication/notification lists are expected to grow large or need animations.
- Add an empty state to `NotificationsFragment` / `HomeFragment` when their lists are empty.
- Consider a lightweight repository class once more than two screens need the same Firebase read/write patterns — right now the duplication is small enough not to be worth it, but it's the first thing to reconsider if a third or fourth screen needs medication CRUD.